

Jisung Yoo

Senior Research Scientist | 3D Computer Vision

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RESEARCH PROFILE

3D vision researcher specializing in large-scale scene reconstruction, neural rendering, and generative 3D for real-world environments. At NAVER LABS, core work includes production-scale Street View 3D and city-scale novel-view synthesis deployed in NAVER Map, with contributions to aerial digital-twin reconstruction. Current research explores 3D Gaussian Splatting, generative video/3D diffusion, and feed-forward 3D representations for extending under-observed scenes while preserving geometric and multi-view consistency. Previous experience includes on-device AR, computational photography, and 3D-content service prototyping for Samsung Galaxy products.

PROFESSIONAL EXPERIENCE

NAVER LABS

Feb 2022 - Present

Senior Research Scientist | Seongnam, South Korea

- Core contributor to the large-scale 3D reconstruction and rendering pipeline behind NAVER Map Street View 3D, enabling spatially consistent 3D visualization and seamless exploration.
- Core contributor to city-scale novel-view synthesis (NVS) for free-viewpoint urban rendering, deployed in NAVER Map FlyingView 3D and media/VFX applications; worked across heterogeneous street-view and aerial/drone imagery.
- Contributed MVS-based dense point-cloud and mesh reconstruction, as well as 3DGS-based scene reconstruction outputs, to city-scale digital-twin pipelines.
- Current research: exploring generative video/3D diffusion and feed-forward 3D Gaussian representations to extend reconstruction and NVS beyond captured viewpoints, emphasizing geometric and multi-view consistency.

Samsung Research

Jan 2021 - Jan 2022

Creative Leader, Team ReMo (Reconstruct Moments) | Seoul, South Korea

- Led an incubation team exploring AR/VR applications and business models for 3D content creation from smartphone photos.
- Prototyped photorealistic 3D-content experiences, including single-image animatable avatars and interactive 3D photos from smartphone image sequences.

Samsung Electronics

Mar 2018 - Dec 2020

Staff Engineer, Mobile Communications Division | Suwon, South Korea

- Developed core time-of-flight (ToF) AR measurement technologies for QuickMeasure, commercialized on the Galaxy S10 5G.
- Developed photo-filter style transfer for My Filters on the Galaxy S20 series; corresponding author of an ECCV 2020 Spotlight paper.
- Implemented the on-device Photo Remaster pipeline for degradation detection, automatic enhancement, and result recommendation, commercialized on the Galaxy S21 series.

- Explored multi-task learning for segmentation and depth estimation. Generated human-depth data using multi-view stereo and synthetic segmentation/depth data using Unreal Engine.

SELECTED PUBLIC WORK AT NAVER LABS

Company-published articles describing public systems and deployments to which I contributed.

[NAVER Map FlyingView 3D - City-scale Novel View Synthesis](#)

Public overview of large-scale NVS using street-view and drone imagery, including multi-sensor 3D alignment, tiled city-scale training/rendering, and treatment of distant/background regions.

[NAVER Map Street View 3D - Large-scale 3D Reconstruction & Visualization](#)

Public overview of the production pipeline from mapping data and precise point clouds to 360 panoramas and depth-aware 3D meshes for seamless spatial navigation and information augmentation.

[Novel View Synthesis for Media & VFX](#)

Public example of real-world urban reconstruction and NVS used to synthesize new viewpoints for screen content, extending the technology beyond mapping.

DOCTORAL RESEARCH PROJECTS

KAIST

Mar 2010 - Feb 2018

Doctoral Research, Electrical Engineering | Daejeon, South Korea

Multi-view Image-based 3D Face Reconstruction

Tricubics Inc.

- Developed preprocessing for eye-blink detection, specular-region detection, and keyframe selection to improve reconstruction quality.
- Implemented visual SLAM and volumetric reconstruction for multi-view 3D face reconstruction.

3D Object and Texture Reconstruction based on Depth and Multi-view Video

National Research Foundation

- Built an RGB-D and turntable-based capture system for depth and multi-view images.
- Reconstructed object geometry using visual hulls refined by depth images, and applied view-dependent texture mapping for photorealistic rendering.

3D Hand Skeleton Estimation based on Proximity Sensor

Samsung Electronics

- Analyzed depth and proximity sensors; estimated 3D hand skeletons from extracted hand-depth data using a randomized decision forest.

3D Video Object Compression Method based on Intermediate Surface

National Research Foundation

- Proposed conversions between 3D video object formats, including voxels and meshes, and developed a compression framework based on the Intermediate Surface.

Layered Panoramic Imaging

Electronics and Telecommunications Research Institute

- Estimated multi-view depth maps and fused them into a panoramic depth map.
- Combined panoramic video with high-resolution regions of interest for interactive viewing.

Multi-object Segmentation and Tracking

Electronics and Telecommunications Research Institute

- Extracted human masks and tracked multiple people in video sequences in real time.

TECHNICAL EXPERTISE

Research: 3D reconstruction, multi-view stereo, RGB-D reconstruction, visual SLAM, neural rendering, novel-view synthesis, 3D Gaussian Splatting, generative video/3D models, feed-forward 3D representations.

Vision & imaging: Image processing, depth estimation, segmentation, image enhancement, style transfer.

Programming & platforms: C/C++, CUDA, OpenGL, Three.js, Android.

PUBLICATIONS

Publication records retained from the August 2023 CV.

International Conference

Filter Style Transfer between Photos

J. Yim, J. Yoo*, W.-J. Do, B. Kim, and J. Choe.
*European Conference on Computer Vision (ECCV), 2020. Spotlight. *Corresponding author.*

International Journals

Pyramid Inter-Attention for High Dynamic Range Imaging

S. Choi, J. Cho, W. Song, J. Choe, J. Yoo, and K. Sohn.
Sensors, 20(18):5102, September 2020.

Image-based object reconstruction using run-length representation

S. S. Hwang, H.-D. Kim, T. Y. Jang, J. Yoo, S. Kim, K. Paeng, and S. D. Kim.
Signal Processing: Image Communication, 51:1-12, February 2017.

Visual Hull-Based Geometric Data Compression of a 3-D Object

Sung Soo Hwang, Wook-Joong Kim, Jisung Yoo, and Seong Dae Kim.
IEEE Transactions on Circuits and Systems for Video Technology, 25(7):1151-1160, July 2015.

Scale-invariant template matching using histogram of dominant gradients

J. Yoo, S. S. Hwang, S. D. Kim, M. S. Ki, and J. Cha.
Pattern Recognition, 47(9):3006-3018, September 2014.

Regional Linear Warping for Image Stitching with Dominant Edge Extraction

J. Yoo, S. S. Hwang, S. D. Kim, M. S. Ki, and J. Cha.
KSII Transactions on Internet and Information Systems, 7(1):2464-2478, October 2013.

EDUCATION

KAIST Mar 2010 - Feb 2018

Ph.D. in Electrical Engineering | Daejeon, South Korea

Dissertation: A study on interactive 3D reconstruction using RGB-D camera

KAIST Mar 2006 - Jan 2010

B.Eng. in Electrical Engineering | Daejeon, South Korea

Gyeonggi Science High School Mar 2004 - Feb 2006

High School Diploma | Suwon, South Korea

PATENTS AND PATENT APPLICATIONS

Identifiers, years, and grant/application categories below are retained from the August 2023 CV; subsequent status changes have not been verified.

Granted patents listed in the source CV**Electronic device and object measurement method thereof**

D. Oh, **J. Yoo**, B. Kim, C. Park, and K. Lee.

U.S. Patent 11,361,461 | 2022

Apparatus and method for creating panorama

M. S. Ki, **J. Yoo**, and S. D. Kim.

U.S. Patent 9,584,723 | 2017

Apparatus and method for generating and consuming three-dimensional (3D) data format to generate realistic panoramic image

M. S. Ki, Y. J. Cho, S. Y. Lim, J. H. Cha, S. K. Beack, Y. J. Lee, J. W. Kim, S. D. Kim, W. J. Kim, **J. Yoo**, and S. S. Hwang.

U.S. Patent 9,185,288 | 2015

Applications listed in the source CV**Electronic apparatus and method for controlling thereof**

J. Yoo, Y. Park, D. Kim, B. Moon, and Y. Lee.

U.S. Patent Application US18/208,976 | 2023

Method for generating image and electronic device therefor

J. Choe, J. Yim, B. Kim, W. Do, **J. Yoo**, K. Lim, and D. Shin.

U.S. Patent Application 17/798,842 | 2023

Method and an electronic device for detecting and removing artifacts/degradations in media

B. Kim, B. Singh, A. S. Shukla, **J. Yoo**, N. G. Pai, P. S. Nair, R. N. Gadde, S. Choi, and S. K. Pasupuleti.

U.S. Patent Application 17/550,751 | 2022

TEACHING EXPERIENCE**Teaching Assistant, KAIST | 2011 - 2012**

Multiple View Geometry for Computer Vision; Matrix Computation.

AWARDS AND HONORS

Samsung Ph.D. Scholarship | Samsung Electronics | Mar 2013 - Feb 2017

Silver Prize, Korean Physics Olympiad (KPhO) | 2005